



Linear Models IV: Multiple Regression

Megan Ayers
 Stat 141 | Fall 2026
 Wednesday, Week 5

1

Goals for Today

- Handling categorical and quantitative explanatory variables at the same time.
- Compare parallel slopes and interaction models for multiple regression
- Transformations

2

Multiple Linear Regression

Form of the Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

How does extending to more predictors change our process?

- What **doesn't** change:
 - Still use **Method of Least Squares** to estimate coefficients
 - Still use `lm()` to fit the model and `predict()` for prediction
- What **does** change:
 - Meaning of the coefficients are more complicated and depend on other variables in the model
 - Need to decide which variables to include and how (linear term, squared term...)

3

Multiple Linear Regression

- We are going to see a few more examples of multiple linear regression this week.
- We will need to return to modeling later in the course once we have learned about statistical inference (i.e., confidence intervals and p-values).

4

Example

Meadowfoam is a plant that grows in the Pacific Northwest and is harvested for its seed oil. In a randomized experiment, researchers at Oregon State University looked at how two light-related factors influenced the **number of flowers per meadowfoam plant**, the primary measure of productivity for this plant. The two light measures were **light intensity** (in $\text{mmol}/\text{m}^2/\text{sec}$) and the **timing of onset of the light** (early or late in terms of photo periodic floral induction).



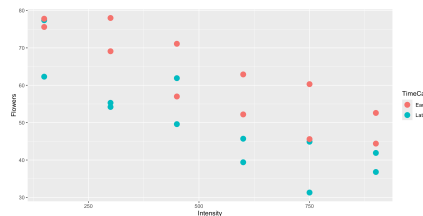
Response variable?

Explanatory variables?

Model Form?

Visualizing the Data

```
1 ggplot(case0901,
2       aes(x = Intensity,
3           y = Flowers,
4           color = TimeCat)) +
5   geom_point(size = 4)
```



- **Q:** How does this plot help us intuit what $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ might be?
- **Side Q:** Why don't I have to include `data =` and `mapping =` in my `ggplot()` layer?

Data Loading and Wrangling

```
1 library(tidyverse)
2 library(Sleuth3)
3 data(case0901)
4
5 # Check out the timing variable
6 count(case0901, Time)

Time  n
1     12
2     12

1 # Recode the timing variable
2 case0901 <- case0901 %>%
3   mutate(TimeCat = case_when(Time == 1 ~ "Late",
4                               Time == 2 ~ "Early"))
5 count(case0901, TimeCat)

TimeCat  n
1   Early 12
2   Late 12
```

Building and Interpreting the Linear Regression Model

Full model form:

```
1 modFlowers <- lm(Flowers ~ Intensity + TimeCat, data = case0901)
2
3 library(moderndiv)
4 get_regression_table(modFlowers)

# A tibble: 3 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept  83.5        3.27      25.5     0       76.7    90.3
2 Intensity  -0.04       0.005     -7.89    0      -0.051 -0.03
3 TimeCat: Late -12.2       2.63     -4.62    0      -17.6   -6.69
```

- Estimated regression line for **TimeCat = "Late"**
- Estimated regression line for **TimeCat = "Early"**:
- Interpreting coefficients?

Building and Interpreting the Linear Regression Model

Full model form:

```
1 modFlowers <- lm(Flowers ~ Intensity + TimeCat, data = case0901)
2
3 library(moderndiver)
4 get_regression_table(modFlowers)

# A tibble: 3 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>    <dbl>    <dbl>   <dbl>    <dbl>    <dbl>
1 intercept  83.5      3.27     25.5     0      76.7     90.3
2 Intensity  -0.04     0.005    -7.89    0     -0.051  -0.03
3 TimeCat: Late -12.2     2.63     -4.62    0     -17.6    -6.69
```

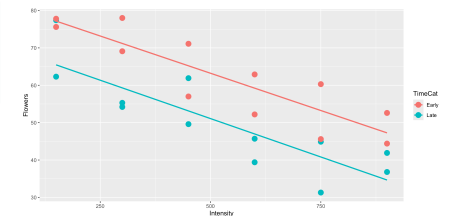
Interpreting coefficients?

- Expect **Flowers** to be 83.5 on average when **Intensity** = 0 and **TimeCat** is early
- Expect **Flowers** to decrease by -0.04 on average when **Intensity** increases by 1, **holding TimeCat constant**
- Expect plants with late **TimeCat** to have -12.2 fewer flowers on average **than those with early TimeCat, holding Intensity constant**

9

Appropriateness of Model Form

```
1 ggplot(case0901,
2       aes(x = Intensity,
3           y = Flowers,
4           color = TimeCat)) +
5   geom_point(size = 4) +
6   geom_smooth(method = "lm", se = FALSE)
```



Is the assumption of **equal slopes** reasonable here?

10

Prediction

```
1 flowersNew <- data.frame(Intensity = c(700, 700), TimeCat = c("Early", "Late"))
2 flowersNew

  Intensity TimeCat
1       700   Early
2       700    Late

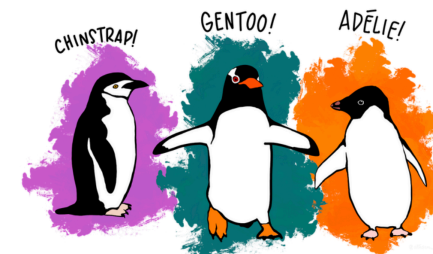
1 predict(modFlowers, newdata = flowersNew)

      1      2
55.13417 42.97583
```

11

Returning to the Palmer Penguins

```
1 library(palmerpenguins)
```



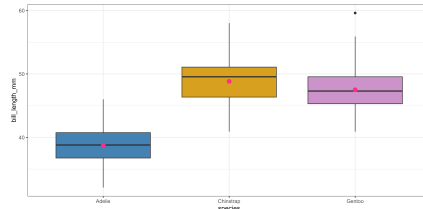
The Palmer Archipelago penguins. Artwork by @allison_horst.

12

Last time: One Categorical Variable

We predicted a penguin's bill length based on their species.

```
1 ggplot(penguins, aes(x = species, y = bill_length_mm, f
2   geom_boxplot() +
3   scale_fill_manual(values = c("steelblue",
4     "goldenrod",
5     "plum3")) +
6   stat_summary(fun = mean,
7     geom = "point",
8     size = 3,
9     color = "deeppink") +
10  guides(fill = "none") +
11  theme_bw()
```



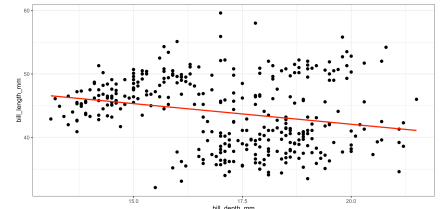
- Pink dots represent the mean value in each group.
- For the single categorical variable model, those pink dots are the predicted values for each group.

13

This time: Add a Quantitative Variable

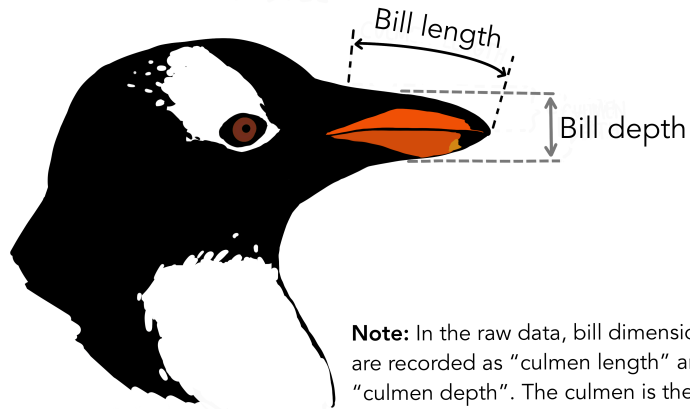
We'll incorporate bill depth for prediction!

```
1 ggplot(penguins, aes(x = bill_depth_mm, y = bill_length
2   geom_point(size = 2) +
3   geom_smooth(method = "lm", se = FALSE, color = "red")
4   theme_bw()
```



- A moderate negative relationship between bill length and bill depth!
- Does this make sense?

14

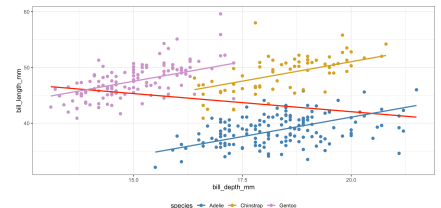


Note: In the raw data, bill dimensions are recorded as "culmen length" and "culmen depth". The culmen is the dorsal ridge atop the bill.

15

What if we include both?

```
1 ggplot(penguins, aes(x = bill_depth_mm,
2   y = bill_length_mm,
3   color = species)) +
4   geom_point(size = 2) +
5   geom_smooth(inherit.aes = FALSE,
6     mapping = aes(x = bill_depth_mm,
7       y = bill_length_mm),
8     method = "lm", se = FALSE,
9     color = "red") +
10  geom_parallel_slopes(se = FALSE) +
11  scale_color_manual(values = c("steelblue",
12    "goldenrod",
13    "plum3")) +
14  theme_bw() +
15  theme(legend.position = "bottom")
```



- **Negative** relationships between bill depth and bill length overall.
- **Positive** relationships between bill depth and bill length when accounting for species!
- **What is going on here??**
- This is a case of **Simpson's Paradox**!
- Side note: this type of graph is really helpful when including both a categorical and quantitative variable in a linear model!

16

Three candidate models

Explanatory variable: **species**

```
1 species_mod <- lm(bill_length_mm ~ species, penguins)
2 get_regression_table(species_mod) %>%
3 select(term, estimate)
```

```
# A tibble: 3 x 2
  term      estimate
  <chr>      <dbl>
1 intercept 38.8
2 species: Chinstrap 10.0
3 species: Gentoo 8.71
```

Explanatory variable: **bill_depth_mm**

```
1 depth_mod <- lm(bill_length_mm ~ bill_depth_mm, penguins)
2 get_regression_table(depth_mod) %>%
3 select(term, estimate)
```

```
# A tibble: 2 x 2
  term      estimate
  <chr>      <dbl>
1 intercept 55.1
2 bill_depth_mm -0.65
```

Explanatory variables: **species** and **bill_depth_mm** (equal slope)

```
1 both_mod <- lm(bill_length_mm ~ bill_depth_mm + species
2 get_regression_table(both_mod) %>%
3 select(term, estimate)
```

```
# A tibble: 4 x 2
  term      estimate
  <chr>      <dbl>
1 intercept 13.2
2 bill_depth_mm 1.39
3 species: Chinstrap 9.94
4 species: Gentoo 13.4
```

- Coefficient interpretations?

03:00

17

Interpreting the Multiple Regression Equation

```
1 get_regression_table(both_mod)
```

```
# A tibble: 4 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 13.2        2.25        5.88     0      8.80    17.6
2 bill_depth_mm 1.39      0.122       11.4     0      1.15    1.63
3 species: Chinstrap 9.94    0.368       27.0     0      9.22   10.7
4 species: Gentoo 13.4      0.512       26.2     0     12.4   14.4
```

$$\hat{y} = 13.2 + 1.39 \cdot x_{\text{Bill Depth}} + 9.94 \cdot x_{\text{Species:Chinstrap}} + 13.4 \cdot x_{\text{Species:Gentoo}}$$

- **Slope Coefficient on Quantitative Variable:** The *expected/predicted* change in response variable, *on average*, per 1 unit increase in the explanatory variable, *holding all other variables constant*.
 - Example: We expect a 1.394mm increase in bill length for every 1mm increase in bill depth, on average, after holding species constant.
- Interpretation on factors from **Categorical Variables** still applies: the *expect/predicted* change in response variable, *on average*, when going from the reference level to the factor's level, *holding all other variables constant*.
 - Example: We expect bill length to be 9.94mm larger on average for Chinstrap penguins compared to Adelie penguins, holding bill depth constant.

18

Interpreting the Multiple Regression Equation

```
1 get_regression_table(both_mod)
```

```
# A tibble: 4 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 13.2        2.25        5.88     0      8.80    17.6
2 bill_depth_mm 1.39      0.122       11.4     0      1.15    1.63
3 species: Chinstrap 9.94    0.368       27.0     0      9.22   10.7
4 species: Gentoo 13.4      0.512       26.2     0     12.4   14.4
```

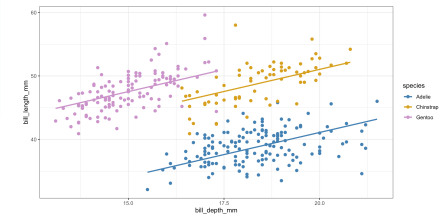
$$\hat{y} = 13.2 + 1.39 \cdot x_{\text{Bill Depth}} + 9.94 \cdot x_{\text{Species:Chinstrap}} + 13.4 \cdot x_{\text{Species:Gentoo}}$$

- **Intercept Coefficient:** The *expected/predicted* value of the response, *on average*, when all explanatory variables are set to 0.
 - What does setting species to 0 mean in this context?

19

Returning to our MLR model - equal slopes

```
1 ggplot(penguins, aes(x = bill_depth_mm, y = bill_length_mm)) +
2   geom_point(size = 2) +
3   geom_parallel_slopes(se = FALSE) +
4   scale_color_manual(values = c("steelblue",
5     "goldenrod",
6     "plum3")) +
7   theme_bw()
```

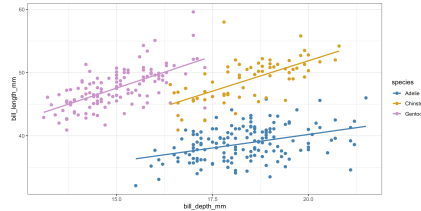


Are equal slopes a reasonable assumption here?

20

Different Slopes Model (ggplot example)

```
1 ggplot(penguins, aes(x = bill_depth_mm, y = bill_length_mm)) +
2   geom_point(size = 2) +
3   geom_smooth(method = "lm", se = FALSE) +
4   scale_color_manual(values = c("steelblue",
5     "goldenrod",
6     "plum3")) +
7   theme_bw()
```



- **Equal slopes** models force the relationship between quantitative predictors and the response variable to be the same for each group in the model.
- In contrast, **different slopes** models allow for **different relationships** between quantitative predictors and the response variable for each group in the model.
- How can we allow our model to have different slopes?

21

Contrasting model forms

Recall the equal slopes model:

$$y = \beta_0 + \beta_1 \cdot X_{\text{Bill Depth}} + \beta_2 \cdot X_{\text{Species:Chinstrap}} + \beta_3 \cdot X_{\text{Species:Gentoo}} + \epsilon$$

How can we allow the slopes to vary?

$$y = \beta_0 + \beta_1 \cdot X_{\text{Bill Depth}} + \beta_2 \cdot X_{\text{Species:Chinstrap}} + \beta_3 \cdot X_{\text{Species:Gentoo}} + \beta_4 \cdot X_{\text{Bill Depth}} \cdot X_{\text{Species:Chinstrap}} + \beta_5 \cdot X_{\text{Bill Depth}} \cdot X_{\text{Species:Gentoo}} + \epsilon$$

- **Coefficient interpretation?**
 - Those involving $X_{\text{Bill Depth}}$ are particularly tricky

22

Different Slopes Model in R

```
1 same_slope <- lm(bill_length_mm ~ bill_depth_mm + species, penguins)
2
3 diff_slope <- lm(bill_length_mm ~ bill_depth_mm * species, penguins)
```

```
1 get_regression_table(same_slope)
```

```
# A tibble: 4 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 13.2        2.25      5.88     0      8.80    17.6
2 bill_depth_mm 1.39      0.122     11.4     0      1.15    1.63
3 species: Chinstrap 9.94      0.368     27.0     0      9.22    10.7
4 species: Gentoo 13.4        0.512     26.2     0     12.4    14.4
```

```
1 get_regression_table(diff_slope)
```

```
# A tibble: 6 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 23.1        3.02      7.65     0     17.1    29.0
2 bill_depth_mm 0.857      0.164      5.22     0     0.534  1.18
3 species: Chinstrap -9.64      5.72     -1.69    0.093  -20.9   1.60
4 species: Gentoo -5.84      4.54     -1.29    0.199  -14.8   3.08
5 bill_depth_mm:speciesC... 1.06      0.31      3.44    0.001  0.455  1.68
6 bill_depth_mm:speciesG... 1.16      0.279      4.17     0     0.615  1.71
```

23

Detour: handling curved relationships

New data context: movie ratings

```
1 library(tidyverse)
2 movies <- read_csv("https://www.lock5stat.com/datasets2e/HollywoodMovies.csv")
3
4 # Restrict our attention to dramas, horrors, and actions
5 movies2 <- movies %>%
6   filter(Genre %in% c("Drama", "Horror", "Action")) %>%
7   drop_na(Genre, AudienceScore, RottenTomatoes)
8 glimpse(movies2)
```

```
Rows: 313
Columns: 16
$ Movie      <chr> "Spider-Man 3", "Transformers", "Pirates of the Carib...
$ LeadStudio <chr> "Sony", "Paramount", "Disney", "Warner Bros", "Warner...
$ RottenTomatoes <dbl> 61, 57, 45, 60, 20, 79, 35, 28, 41, 71, 95, 42, 18, 2...
$ AudienceScore <dbl> 54, 89, 74, 90, 68, 86, 55, 56, 81, 52, 84, 55, 70, 6...
$ Story      <chr> "Metamorphosis", "Monster Force", "Rescue", "Sacrific...
$ Genre      <chr> "Action", "Action", "Action", "Action", "Action", "Ac...
$ TheatersOpenWeek <dbl> 4252, 4011, 4362, 3103, 3778, 3408, 3959, 3619, 2911,...
$ OpeningWeekend <dbl> 151.1, 70.5, 114.7, 70.9, 49.1, 33.4, 58.0, 45.3, 19...
$ BOAvgOpenWeekend <dbl> 35540, 17577, 26302, 22844, 12996, 9791, 14663, 12541...
$ DomesticGross <dbl> 336.53, 319.25, 309.42, 210.61, 140.13, 134.53, 131.9...
$ ForeignGross <dbl> 554.34, 390.46, 654.00, 245.45, 117.90, 249.00, 157.1...
$ WorldGross <dbl> 890.87, 709.71, 963.42, 456.07, 258.02, 383.53, 289.0...
```

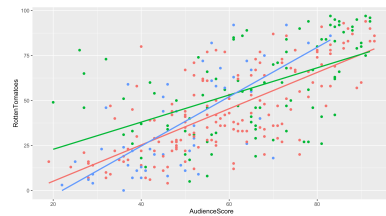
Response variable: RottenTomatoes

Explanatory variables: AudienceScore

24

Exploring the Data

```
1 ggplot(data = movies2,
2       mapping = aes(x = AudienceScore,
3                     y = RottenTomatoes,
4                     color = Genre)) +
5   geom_point() +
6   geom_smooth(method = "lm", se = FALSE)
```

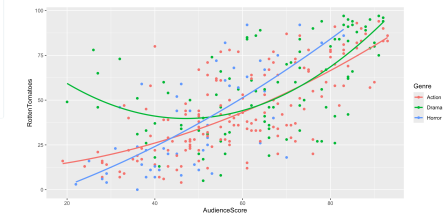


- Q: What do the trends suggest, would it make sense to include interaction terms in the model?
- Q: Does anyone spot a curved relationship for one of the genres?

25

Adding a Curve to your Scatterplot

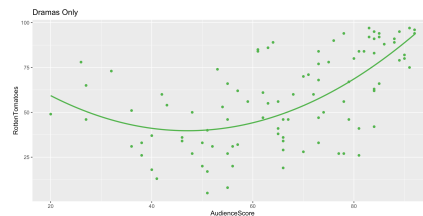
```
1 ggplot(data = movies2,
2       mapping = aes(x = AudienceScore,
3                     y = RottenTomatoes,
4                     color = Genre)) +
5   geom_point() +
6   geom_smooth(method = "lm", se = FALSE,
7             formula = y ~ poly(x, degree = 2))
```



26

Adding a Curve to your Scatterplot

```
1 # Filtering to only Dramas
2 dramas <- movies2 %>% filter(Genre == "Drama")
3
4 ggplot(data = dramas,
5       mapping = aes(x = AudienceScore,
6                     y = RottenTomatoes)) +
7   geom_point(color = "#55B84F") +
8   geom_smooth(method = "lm", se = FALSE,
9             color = "#55B84F",
10            formula = y ~ poly(x, degree = 2)) +
11   ggtitle("Dramas Only")
```



27

Using Transformations to Account for Non-Linear Relationships

```
1 # Adding a column for AudienceScore^2
2 dramas <- dramas %>%
3   mutate(AudienceScoreSq = AudienceScore^2)
4
5 ggplot(data = dramas,
6       mapping = aes(x = AudienceScoreSq,
7                     y = RottenTomatoes)) +
8   geom_point(color = "#55B84F") +
9   geom_smooth(method = "lm", se = FALSE,
10             color = "#55B84F") +
11   ggtitle("Dramas Only")
```



- Notice that the relationship between AudienceScore^2 and RottenTomatoes is approximately linear.
- AudienceScore^2 is a transformation of AudienceScore

28

Fitting a Model with a Transformed Variable

```
1 dramas <- movies2 %>% filter(Genre == "Drama")
2 mod2 <- lm(RottenTomatoes ~ poly(AudienceScore, degree = 2, raw = TRUE),
3           data = dramas)
4 get_regression_table(mod2)
```

A tibble: 3 × 7

term	estimate	std_error	statistic	p_value	lower_ci	upper_ci
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 intercept	98.8	21.3	4.64	0	56.6	141.
2 poly(AudienceScore, de...	-2.51	0.722	-3.48	0.001	-3.94	-1.08
3 poly(AudienceScore, de...	0.027	0.006	4.58	0	0.015	0.038

- We'll practice a very common transformation in HW 5 - the **log transformation**
- **Q**: Why is it still called a *linear* regression if the model also handles curved relationships?
- Transformations are super powerful: we can now use linear models even if the *untransformed* variables have non-linear relationships.

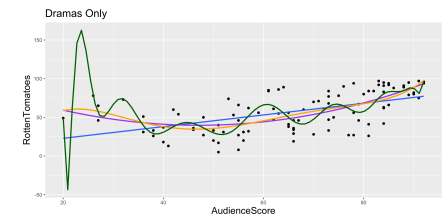
29

Model Building Guidance

What degree of polynomial should I include in my model?

Guiding Principle: Capture the general trend, not the noise.

$$y = f(x) + \epsilon$$
$$y = \text{TREND} + \text{NOISE}$$



30

Recap

- We've now looked at linear models that combine quantitative and categorical explanatory variables
- They can be complicated to interpret - review the examples here as you practice!
- We've compared **fixed and varying slope models**, and seen how to visualize each in **ggplot2** (**geom_parallel_slopes()** vs regular **geom_smooth()**)
- We've seen how **transformations** can help us proceed with linear models even when our data appear to have nonlinear relationships

Next time

- Linear regression with many variables
- Multicollinearity
- Model selection

31